

SAB Mockup Features — Design Document

Date: 2026-03-05 **Origin:** Alliance SAB Meeting mockup presentation (Feb 2022, Jennifer Smith) **Approach:** Full-stack vertical slices — each feature implemented end-to-end (pipeline → API → WebUI) before moving to the next.

Scope

Four features, in priority order:

1. Variant sequence display (mutant protein rows in alignment)
2. Protein domains track (InterPro/UniProt annotations)
3. Disease/phenotype tables (comparison matrix + phenotype display)
4. Guide tree capture + display (Clustal Omega guide tree as dendrogram)

Current State Summary

Capability	Status
MSA viewer with Clustal coloring	Implemented (Nightingale)
Conservation track/coloring	Implemented (linegraph + MSA overlay)
Variant position highlighting	Implemented (amber overlays + shape-coded track)
Mutant protein rows	Gap — alt sequences produced by pipeline but not identified in UI
Protein domains	Gap — no InterPro/UniProt integration
Disease/phenotype annotations	Gap — no infrastructure
Guide tree	Gap — computed by clustalo but discarded
PositionInfoPanel	Built but orphaned (never wired into any viewer)
AlignmentSearch	Built but orphaned
VisualizationToolbar	Built but orphaned

Feature 1: Variant Sequence Display

Background

The seq_retrieval pipeline already produces alternative protein sequences as separate FASTA entries (e.g., >TP53_NM_011640.3_alt1). These go into the Clustal Omega alignment and appear as rows in the output. The

`aligned_seq_info.json` contains `embedded_variants` with alignment positions. The WebUI does not currently distinguish alt sequence rows from reference rows.

Pipeline Changes (`seq_retrieval`)

Add fields to the `SeqInfo` entries in `aligned_seq_info.json` for alt sequence entries:

- `is_alt_sequence`: `bool` — flag to identify alt rows
- `parent_sequence_name`: `string` — links alt row back to its reference sequence
- `variant_summary`: `string` — human-readable summary (e.g., “p.Leu278Ser”)

These fields are added in `seq_retrieval.py` when constructing `alt_info` objects.

WebUI Changes (`VirtualizedAlignment`)

- Parse alt sequence rows using the `is_alt_sequence` flag from seq info
- Group alt rows visually below their parent reference row
- Render alt rows with distinct styling:
 - Indented row label
 - Colored left border (red/salmon, consistent with mockup mv1/mv2 styling)
 - Variant position highlighted in the sequence
- Wire up the existing orphaned `PositionInfoPanel` component:
 - Show on click/hover of a variant position
 - Display: variant ID, HGVS notation, ref→alt change, genomic coordinates, substitution type

Data Flow

Pipeline: ref FASTA + alt FASTA → Clustal Omega → `alignment.aln` + `aligned_seq_info.json`
(with `is_alt_sequence`, `parent_sequence_name`)

WebUI: Parse alignment → identify alt rows via seq info → render grouped with distinct styling

Feature 2: Protein Domains Track

Data Source

InterPro REST API: <https://www.ebi.ac.uk/interpro/api/entry/interpro/protein/UniProt/{accession}>

Provides rich domain data including Pfam, PROSITE, CDD, grouped by type (Domain, Family, Homologous Superfamily). Fallback to UniProt Features API if InterPro is unavailable.

Pipeline Changes (seq_retrieval)

- After sequence retrieval, map each protein to its UniProt accession:
 - Via NCBI RefSeq → UniProt mapping (UniProt ID mapping API)
 - Or from Alliance API gene data which includes cross-references
- Fetch InterPro domain annotations per protein
- Map domain coordinates from original protein positions to alignment positions (reuse the same position mapping used for variants in `_add_alignment_positions`)
- Store in `aligned_seq_info.json` as a new `domains` field per sequence:

```
{
  "domains": [
    {
      "accession": "IPR009057",
      "name": "Homeobox Domain",
      "source": "Pfam:PF00046",
      "type": "domain",
      "start": 98,
      "end": 152,
      "alignment_start": 120,
      "alignment_end": 178
    }
  ]
}
```

WebUI Changes

- New `DomainTrack` component using `nightingale-track` with rectangle shape features
- Render one track per unique domain, colored consistently across species (same domain = same color)
- Domain legend below the alignment with colored boxes and labels (matching mockup style: “Homeobox Domain”, “Fictitious 1 Domain”)
- Toggle visibility via the existing “Show” overlay checkbox panel (add “Protein Domains” checkbox)
- Domains rendered as semi-transparent colored bands spanning the alignment columns

Feature 3: Disease/Phenotype Tables

Data Source

Alliance of Genome Resources API — fetched during pipeline processing and included in result JSON.

Pipeline Changes

For each variant ID in the job, fetch enriched annotations from the Alliance API:

- Variant details: HGVS notation, molecular consequence, type
- Disease annotations from associated alleles
- Phenotype annotations from associated alleles

Store as enriched variant data in `aligned_seq_info.json`:

```
{
  "embedded_variants": [{
    "variant_id": "...",
    "hgvs_notation": "NP_000316.2:p.Leu278Ser",
    "molecular_consequence": "missense_variant",
    "diseases": [
      {
        "do_id": "DOID:0110048",
        "name": "Axenfeld-Rieger syndrome",
        "source": "ClinVar via RGD",
        "evidence_code": "IAGP",
        "references": ["PMID:12624268"]
      }
    ],
    "phenotypes": [
      {
        "term": "abnormal eye pressure",
        "source": "MGI",
        "evidence_code": "TAS",
        "allele_symbol": "Pitx2<tm1Abc>",
        "references": ["PMID:18424556"]
      }
    ]
  }]
}
```

WebUI — Disease Comparison Matrix

New `DiseaseComparisonMatrix` component:

- Rows = Disease Ontology terms
- Columns = variants grouped by species
- Cells colored when association exists:
 - Blue = direct annotation
 - Orange = inferred from multi-variant allele (as in mockup)
- Expandable rows for disease hierarchy (e.g., “Axenfeld-Rieger syndrome” → subtypes)
- Toggle: “Include annotations from multi-variant alleles”

- Accessible from a “Show Disease/Phenotype Comparison” button on the result page

WebUI — Phenotype Display Table

New `PhenotypeTable` component:

- Columns: Variant, Species, Associated Allele, Phenotype, Evidence, Source, References
- Filterable by variant and species
- Sortable columns
- Reference links to PubMed (PMID external links)

Placement

New tabs or expandable sections below the alignment viewer on the result page.

Feature 4: Guide Tree Capture + Display

Pipeline Changes

Add `--guidetree-out=guidetree.nwk` to all clustalo invocations:

- `pipeline_components/alignment/scripts/alignment_wrapper.sh` (production/Step Functions)
- `api/src/local_pipeline.py` (local EC2 path)
- `pipeline_components/alignment/protein-msa.nf` (legacy Nextflow)

Copy `guidetree.nwk` alongside `alignment-output.aln` in results output.

API Changes

New endpoint: `GET /api/pipeline-job/{uuid}/result/tree`

Returns the Newick-format guide tree file.

WebUI Changes

- Fetch tree from the new endpoint alongside alignment and seq-info results
- New `PhylogeneticTree` component:
 - Parse Newick format (lightweight parser, e.g., `newick-js` npm package or custom — format is simple)
 - Render as horizontal SVG dendrogram/cladogram
 - Position to the left of the alignment (like PANTHER) or as a collapsible side panel
 - Species labels match the MSA row labels
 - Sequence order in the alignment should match tree leaf order for visual consistency

Scope Limits

- Display only — no tree editing or rerooting
 - No bootstrap values (Clustal Omega guide trees don't produce them)
 - Optional toggle ("Show Tree" checkbox in the overlay panel)
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Orphaned Components to Wire Up

During implementation, the following existing but disconnected components should be integrated:

Component	Current State	Integration Target
PositionInfoPanel	Built, never imported	Wire into VirtualizedAlignment for variant click/hover
AlignmentSearch	Built, never imported	Add to toolbar for sequence/position search
VisualizationToolbar	Built, never imported	Evaluate replacing current ad-hoc toolbar controls

Feature 5: Color Scheme Legend

Background

Users have requested an explanation of what the alignment colors mean. The viewer supports 16+ color schemes (Clustal2, Hydrophobicity, Taylor, Zappo, etc.) but there is no visual legend explaining the mapping.

WebUI Changes

- New `ColorSchemeLegend` component:
 - Displays a color key for the currently active color scheme
 - Shows amino acid → color mappings as colored swatches with residue labels
 - For property-based schemes (Hydrophobicity, Charged, Polar), group by property category
 - Collapsible/expandable panel below the color scheme dropdown
 - Updates dynamically when the user switches color schemes
- Position: adjacent to or below the existing color scheme dropdown in the toolbar

No Pipeline/API Changes Required

Color scheme definitions are entirely client-side (in the Nightingale MSA component and/or local config).

Implementation Order

1. **Variant sequence display** — Lowest risk, alt sequences already in alignment output. Pipeline: add metadata fields. WebUI: styling + PositionInfoPanel wiring.
 2. **Color scheme legend** — WebUI-only, no pipeline changes. Quick win for user experience.
 3. **Guide tree** — Small scope, well-defined. Pipeline: one flag addition. API: one new endpoint. WebUI: SVG tree component.
 4. **Protein domains** — Medium scope. Pipeline: InterPro API integration + position mapping. WebUI: Nightingale track + legend.
 5. **Disease/phenotype** — Largest scope. Pipeline: Alliance API integration. WebUI: two new table components.
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Out of Scope (Future Work)

- Variant filtering by type/consequence/position/domain (from mockup slide 13)
- Isoform selection at result-display time (currently submit-time only)
- Logarithmic variant density bar graph (InterPro/JBrowse style, mockup slides 22-23)
- Per-ortholog “show variants” / “show protein” toggles at result time
- Exon boundary markers (data flows through pipeline but not visualized — partially related to Feature 1)
- Export enhancements (FASTA/PNG/PDF via VisualizationToolbar)